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Section: H

Course: Software Engineering Lab 1

Problem Statement #18: Clinical Trial Subject Recruitment Engine

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall evaluate candidate survey responses against trial protocol inclusion and exclusion criteria and classify eligibility status.	High	Pass: ineligible candidate flagged with the specific failed criteria. Fail: candidate with an exclusion criterion marked eligible.	Core screening logic of the recruitment engine.
FR-002	Functional	The system shall allow a Trial Candidate to register an account and submit a pre-screening survey form.	High	A new candidate can sign up, log in and submit the survey. Submitted responses are stored and visible to the Clinical Researcher.	Needed to collect candidate data for eligibility checks.
FR-003	Functional	The system shall record informed consent from the Trial	High	Consent is captured with a digital signature and	Required for ethical and regulatory compliance.

		Candidate with a timestamp before any trial visit data is captured.		timestamp. If consent is not given, the candidate cannot proceed to the visit stage.	
FR-004	Functional	The system shall allow the Clinical Researcher to schedule study visits for each enrolled subject and log visit compliance.	Medium	Researcher can create, view and update visit entries. Each entry stores date, visit type and completion status.	Supports visit compliance tracking.
FR-005	Functional	The system shall generate a subject enrollment report that lists eligible candidates, consent status and visit completion.	Medium	Report can be exported as PDF or CSV and includes all three fields for every subject.	Gives the researcher a single view of trial progress.
NFR-001	Non-Functional	The recruitment database shall comply with 21 CFR Part 11 electronic record standards including cryptographic audit trails.	High	Benchmarking tests confirm target latency and security standards under simulated peak load. Audit log entries are hash chained.	Regulatory requirement for clinical data systems.
NFR-002	Non-Functional	The system shall respond to eligibility	Medium	Load test with 200 concurrent users shows	Keeps the pre-screening flow usable

		evaluation requests within 3 seconds for up to 200 concurrent users.		95th percentile response time under 3 seconds.	during recruitment drives.